



Prop. Let $V = \{f: [-1,1] \rightarrow \mathbb{R} \mid f \text{ is continuous}\}$ with equipped an inner product

$$\langle f, g \rangle = \int_{-1}^1 f(t) \cdot g(t) dt$$

Let W be the subspace of odd functions. Find $W^\perp$.

Proof. $f \in W^\perp \Leftrightarrow$ for any odd function g , $\int_{-1}^1 f(t) g(t) dt = 0$

Since for any even function g and any odd f , fg is odd, so $\int_{-1}^1 fg = 0$.
That is, $W^\perp$ contains all even function. Moreover, for any $f \in V$,

$$f(x) = \underbrace{\frac{f(x) + f(-x)}{2}}_{\text{even}} + \underbrace{\frac{f(x) - f(-x)}{2}}_{\text{odd}}$$

So, for subset W' of V consisting by all even functions,

$$V = W + W', \text{ and } W' \subseteq W^\perp.$$

Now, clearly $W \cap W' = \{0\}$, $V = W \oplus W'$.

If $W' \neq W^\perp$, then, for some $h \in W^\perp$, $h \notin W'$. Again, h can be decomposed as

$$h(x) = \underbrace{\frac{h(x) + h(-x)}{2}}_{\in W'} + \underbrace{\frac{h(x) - h(-x)}{2}}_{\in W}$$

Since $W' \subset W^\perp$, $h(x) - \frac{h(x) + h(-x)}{2} \in W^\perp$, and $\in W$.

But, this implies that it is contained in $W \cap W^\perp$, so it is zero:

$$\text{If } x \in W \cap W^\perp, \text{ then } \langle x, x \rangle = 0, \text{ or } x = 0.$$

Thus, $h(x) - \frac{h(x) + h(-x)}{2} = \frac{h(x) - h(-x)}{2} = 0$, so $h(x) = h(-x)$, even. Contradicts to $h \notin W'$.

Now, we can construct a projection of V onto W as:

$$P: V \rightarrow V: f(x) \mapsto \frac{f(x) - f(-x)}{2}$$